

Temporal Fairness in Educational Early-Warning Systems

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ABSTRACT

Machine learning-based early-warning systems are used in higher education to identify students at risk of failure or withdrawal. However, fairness is often evaluated only after all course information is available, whereas interventions must occur early to be useful. This study examines temporal fairness using the Open University Learning Analytics Dataset (OULAD). We compare an early-course setting using the first 25% of course activity with a full-course setting using all available information. Logistic Regression, Random Forest, and XGBoost are evaluated for predictive performance and subgroup fairness across gender, age band, and IMD (deprivation) band. We focus on false negative rate gaps, because false negatives correspond to at-risk students who are not flagged for support. Results show that performance improves substantially over time; XGBoost ROC-AUC increases from 0.895 to 0.986. FNR gaps also shrink, with the IMD-band gap decreasing from 0.197 to 0.056 for XGBoost. SHAP and PFI analyses suggest that predictions are mainly driven by assessment and VLE engagement features rather than direct demographic features.

KEYWORDS

Machine Learning, Fairness, Explainability, Educational Data Mining; Algorithmic Fairness; Temporal Fairness; Early-Warning Systems; Explainable AI; Learning Analytics; Open University Learning Analytics Dataset (OULAD)

1 INTRODUCTION

The integration of predictive analytics into tertiary education has led to the widespread adoption of Early-Warning Systems (EWS) designed to identify students vulnerable to academic failure or attrition [13]. By leveraging complex behavioral, demographic, and historical data captured within Virtual Learning Environments (VLE), these systems provide institutional agents with actionable insights to deploy resource-intensive interventions [7]. While the primary optimization objective of these frameworks is maximizing predictive performance—typically quantified through accuracy or the Area Under the Receiver Operating Characteristic curve (AUROC)—a critical, yet frequently secondary, consideration is the equity of the downstream allocations they inform [6].

As machine learning models are increasingly entrusted with gate-keeping institutional support, algorithmic fairness has emerged as a foundational area of inquiry within the Educational Data Mining (EDM) and Learning Analytics and Knowledge (LAK) communities [10]. Historical biases embedded within administrative data structures—such as grading variations across socioeconomic brackets, uneven access to digital resources, and structural barriers faced by minoritized groups—can be systematically learned, amplified, and legitimized by predictive models [14]. If a predictive framework

consistently misclassifies at-risk students belonging to a vulnerable demographic subgroup as safe, those students are systematically denied essential academic interventions.

To formalize these equity considerations, algorithmic fairness research has established various statistical benchmarks, including demographic parity, equalized odds, and predictive parity [2]. However, the existing literature predominantly treats these criteria through a *static* lens. Fairness audits are conventionally executed post-hoc, utilizing a complete, static snapshot of the academic term. This standard paradigm fails to align with the temporal execution of actual early-warning mechanisms. In practice, an EWS operates within an evolving data stream, generating rolling predictions from incomplete temporal windows (e.g., at week 2, week 4, or week 8 of a term) when behavioral signals are noisy and sparse.

This operational reality introduces the challenge of *Temporal Fairness*. The predictive behavior and fairness profiles of machine learning models fluctuate dynamically as a function of elapsed time and data accumulation. A model that achieves statistical parity when evaluated on the full-course dataset may exhibit extreme performance disparities during the critical early stages of the term precisely when academic interventions are most effective. Conversely, early-course predictions are highly vulnerable to False Negatives because vulnerable students may not have accumulated sufficient VLE interaction logs to trigger an automated warning signal, which may delay the point at which these students are classified as at risk.

This study provides a rigorous diagnostic evaluation of temporal fairness dynamics within an EWS framework. Focusing on False Negative Rates (FNR) across demographic subgroups, we analyze how predictive equity evolves as a student moves from early-stage data environments to full-course completeness. We leverage the Open University Learning Analytics Dataset (OULAD) [11] to address the following three interconnected Research Questions (RQs):

- **RQ1:** How do subgroup fairness disparities, specifically measured via False Negative Rate gaps (ΔFNR), evolve dynamically between early-course prediction windows (25% temporal checkpoint) and full-course conclusions (100% temporal checkpoint)?
- **RQ2:** To what degree does algorithmic complexity (comparing interpretable Logistic Regression against non-linear ensemble frameworks like Random Forest and XGBoost) exacerbate or mitigate temporal fairness disparities under highly volatile, early-stage data constraints?
- **RQ3:** Can post-hoc explainability methods, specifically Tree-based Shapley Additive exPlanations (TreeSHAP), diagnose whether observed early temporal fairness gaps are driven by distinct behavioral differences across subgroups or by the model utilizing demographic proxy variables within the feature space?

By addressing these questions, this paper seeks to move the EDM field past static equity evaluations and toward a temporally aware paradigm of fair predictive analytics.

2 RELATED WORK

2.1 Educational early-warning systems and learning analytics

Learning analytics and educational data mining have long used institutional records and digital trace data to identify students at risk of failure or dropout [7, 13]. In online and blended learning environments, Virtual Learning Environment (VLE) logs are especially useful because they provide time-stamped evidence of engagement, such as access to course materials, forum participation, and assessment-related activity. The Open University Learning Analytics Dataset (OULAD) was designed for this type of research and combines demographic information, assessment records, final outcomes, and VLE clickstream data across multiple course presentations [11]. Prior work on early-warning systems typically evaluates whether risk can be predicted accurately before the course ends. Our work builds on this setting, but focuses on the timing of reliability: an early-warning model is only useful for intervention if it identifies at-risk students early enough and does so consistently across subgroups.

2.2 Algorithmic fairness in education

Algorithmic fairness is increasingly important in education because predictive models can influence the allocation of academic support, advising, or other institutional resources [10]. In learning analytics, unfairness is often evaluated through subgroup differences in model performance, sometimes called slicing analysis [6]. This perspective is relevant for early-warning systems because the most harmful error is often a false negative: a student who will fail or withdraw is predicted as not at risk, and therefore may not receive timely support. General fairness definitions such as equal opportunity and equalized odds formalize such error-rate disparities across groups [2, 9]. Educational fairness work also cautions that removing demographic attributes does not automatically remove unfairness, because other features may act as proxies or may reflect structural inequalities [14]. In this project, we therefore distinguish between sensitive attributes used for fairness auditing (gender, age band, and IMD (deprivation) band) and educational background features, such as highest prior education, which may be predictive but are not treated as sensitive attributes in our fairness evaluation.

2.3 Explainability for predictive student models

Because early-warning systems may inform high-stakes educational interventions, predictive performance alone is insufficient. Explainability methods can help diagnose which features drive predictions and whether model behaviour changes across course stages. SHAP assigns feature attribution values to individual predictions and provides both local and global explanations of model behaviour [12]. We use SHAP and feature-family importance to compare whether early and full-course models rely mainly on assessment performance, VLE engagement, or static background variables. We also use permutation feature importance (PFI) to measure

how much model performance degrades when individual features are shuffled [1], and partial dependence plots (PDP) and individual conditional expectation (ICE) plots to visualize learned feature effects [5, 8]. These tools are used diagnostically: they explain the trained models, but they do not establish causal mechanisms behind student outcomes.

3 DATASET

We use the Open University Learning Analytics Dataset (OULAD), a public dataset containing information about distance-learning courses at The Open University. OULAD contains records for 32,593 students across 22 course presentations, together with assessment results and more than 10 million daily VLE interaction records. The dataset is appropriate for this project because it combines final academic outcomes with time-dependent learning activity and demographic or socioeconomic attributes, allowing us to study both prediction timing and subgroup fairness.

OULAD is provided as seven linked CSV files. We use `studentInfo.csv` as the base student-level table because it contains the final course outcome and demographic variables. We merge `courses.csv` to obtain the length of each course presentation, which is required for the temporal cutoff. We use `studentRegistration.csv` for registration timing, but deliberately exclude `date_unregistration` because it would directly leak withdrawal information. Assessment features are constructed by merging `assessments.csv` with `studentAssessment.csv`; from these files we aggregate the number of submitted assessments, mean, minimum, and maximum scores, weighted score sums, submitted assessment weight, late submission rate, and average submission timing. VLE engagement features are constructed by merging `studentVle.csv` with `vle.csv`; we aggregate total clicks, active days, click records, clicks per active day, and clicks by activity type.

The prediction target is a binary at-risk label. Students with final outcomes `Fail` or `Withdrawn` are labelled as at risk ($y = 1$), while students with `Pass` or `Distinction` are labelled as not at risk ($y = 0$). This target matches the intended use of an educational early-warning system: students predicted as at risk could be considered for additional support. The binary target is approximately balanced, with 17,208 not-at-risk cases (52.8%) and 15,385 at-risk cases (47.2%).

To study temporal fairness, we construct two versions of the modelling table. The `early_25pct` setting uses only information available in the first 25% of each course presentation. For each student-course instance, VLE logs and assessment records are included only if their date is less than or equal to 0.25 times the corresponding course length. The `full_100pct` setting uses all available course information and serves as an upper-bound benchmark rather than a realistic early intervention point. Comparing these two settings allows us to examine whether subgroup reliability improves as more behavioural and assessment evidence becomes available.

For fairness auditing, we analyse three subgroup attributes: gender, age_band, and imd_band. IMD (deprivation) band is an area-level proxy for socioeconomic deprivation and is therefore treated as a socioeconomic fairness attribute. In line with feedback received during the project update, `highest_education` is retained as a predictive educational background feature but is not treated

as a sensitive attribute for fairness evaluation. This distinction is important because prior education is semantically related to academic performance and may be legitimately useful for prediction, while still being informative in the explainability analysis.

Before model training, categorical variables are one-hot encoded and numerical variables are imputed with median values. Missing engineered activity features are set to zero, representing no observed activity or no submitted assessment before the relevant cutoff. We use a stratified train-test split and apply the same student-course instances to the early and full-course experiments, so differences between temporal stages reflect the available feature information rather than different test populations.

4 METHODS

We developed a modular experimental design to evaluate how predictive performance, statistical fairness, and model explainability change across the two temporal stages. The temporal data environments are derived from OULAD using course-length-based cutoffs, ensuring that early-course and full-course models differ only in the information available at prediction time.

We selected Logistic Regression, Random Forest, and XGBoost because they represent increasing levels of model complexity for tabular classification. Logistic Regression serves as a simple and interpretable linear baseline, allowing us to assess whether more complex models provide meaningful improvements. Random Forest captures non-linear relationships and feature interactions through an ensemble of decision trees, while reducing the instability of individual trees. XGBoost provides a high-performing gradient-boosted tree benchmark that is commonly effective on structured tabular data. Comparing these three models allows us to examine whether stronger predictive performance also leads to smaller, larger, or differently distributed subgroup fairness gaps.

Model architecture is described in more detail in Section 5.1. The specific fairness and explainability methods are described in the following subsections.

4.1 Fairness

4.1.1 Performance metrics. We evaluate global predictive performance using accuracy, balanced accuracy, precision, recall, F1-score, and ROC-AUC. Because the task is to identify at-risk students, recall is especially important: low recall means that students who fail or withdraw are not flagged for support.

4.1.2 Error measurements. Beyond global performance, we analyse error rates for the at-risk class. The False Negative Rate (FNR) measures the proportion of truly at-risk students that the model incorrectly predicts as not at risk:

$$\text{FNR} = \frac{\text{FN}}{\text{FN} + \text{TP}}.$$

This is the most important error metric in our setting, because a false negative means that a student who may need support is not identified by the system. We also report the False Positive Rate (FPR), which captures unnecessary interventions directed at students who are not at risk, and Recall or True Positive Rate (TPR), where $\text{TPR} = 1 - \text{FNR}$.

4.1.3 Sensitive subgroup analysis. Fairness is also calculated and quantified at the level of subgroups based on a set of sensitive features available in OULAD: gender, age_band, and imd_band. The Index of Multiple Deprivation (IMD) band is an indicator of area-level socioeconomic status. Our primary fairness criterion is the FNR gap (ΔFNR), defined as the absolute difference in FNR between the best and worst performing demographic subgroup within each attribute.

4.2 Explainability methods

We use SHAP to quantify local and global feature contributions, reporting mean absolute SHAP values across the test set. For tree-based models, we use TreeSHAP; for Logistic Regression, we use LinearSHAP. We complement SHAP with permutation feature importance (PFI), which measures the drop in balanced accuracy after shuffling each feature. PDP and ICE plots were used as supplementary diagnostics to inspect feature effects and individual heterogeneity, but the main report focuses on SHAP and PFI because they directly support our temporal comparison. Implementation details are available in the source code.¹

4.3 Models

To assess whether model complexity exacerbates or mitigates temporal fairness disparities, we selected three classifiers that span different levels of complexity while remaining appropriate for structured tabular classification.

4.3.1 Logistic Regression. is an intrinsically interpretable linear baseline model. The decision boundary of this model is a weighted linear combination of input features:

$$P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}} \quad (1)$$

Although Logistic Regression's feature attributions are directly readable from its learned coefficients, we nevertheless apply LinearSHAP in the explainability phase. This keeps the attribution methodology consistent across all three models, and LinearSHAP exploits the same linearity assumption to produce exact Shapley values without approximation overhead.

4.3.2 Random Forest. [3] is a bagged ensemble of decision trees where each decision tree is trained on a sample of the data and predictions are dictated by a majority vote:

$$\hat{y} = \text{mode}\{h_t(\mathbf{x})\}_{t=1}^T \quad (2)$$

where $h_t(\mathbf{x})$ is the prediction of the t -th decision tree, T is the tree count, and $\text{mode}(\cdot)$ is majority predicted class. An important property of Random Forests is that as individual trees grow deeper with more available data, the marginal contribution of any single feature becomes distributed across many splits, which dampens the magnitude of individual feature attributions. This is empirically reflected in Figure 4, where Random Forest consistently produces lower mean SHAP magnitudes than Logistic Regression and XGBoost for many top-ranked features, despite achieving competitive predictive performance. This should be considered when comparing SHAP magnitudes across models.

¹Explainability source code: <https://github.com/Vlad-Andres/HCML-project/blob/main/src/explainability.py>

4.3.3 *XGBoost*. [4] is a gradient-boosted tree classifier that builds trees sequentially, with each new tree correcting residual errors from the previous ensemble:

$$\hat{y} = \sum_{t=1}^T \eta \cdot h_t(\mathbf{x}) \quad (3)$$

where h_t denotes the t -th decision tree, T the number of trees in the ensemble, and η the learning rate. As the most complex of the three models, XGBoost can capture non-linear feature interactions in structured tabular data. It serves as a high-performing benchmark and allows us to test whether stronger predictive performance corresponds to smaller or larger subgroup fairness gaps.

5 EXPERIMENTAL SETUP

All experiments use the same stratified train-test split across the early-course and full-course datasets, ensuring that temporal comparisons reflect differences in available information rather than differences in test populations. Each model is trained separately at the early_25pct and full_100pct stages, resulting in six trained models in total. Predictive performance is evaluated using accuracy, balanced accuracy, precision, recall, F1-score, and ROC-AUC. Fairness is evaluated through subgroup-level selection-rate gaps and FNR gaps across gender, age_band, and imd_band. Explainability is assessed using SHAP and PFI, with PDP/ICE used only as supplementary diagnostics.

5.1 Model Architecture

Figure 1 summarizes the project pipeline. We first merge the relevant OULAD tables into a student-course modelling table, define the binary at-risk target, and construct two temporal feature sets: early_25pct and full_100pct. The same three classifiers are then trained separately at both temporal stages, followed by performance, fairness, and explainability analyses. For interpretation, engineered variables are grouped into five feature families: VLE engagement, Assessment, Demographic, Course context, and Registration/other. These families are used only for grouped explanation analysis and do not change the underlying model inputs.²

6 RESULTS

This section presents the predictive performance, fairness analysis and explainability of the proposed Early Warning System using the Open University Learning Analytics Dataset (OULAD). To investigate *Partial Temporal Fairness*, all experiments were conducted at two prediction stages:

- **Early-course prediction (early_25pct):** using only the first 25% of available student activity.
- **Full-course prediction (full_100pct):** using the complete course history.

Three classifiers, namely Logistic Regression, Random Forest, and XGBoost, were trained and evaluated at both stages. We first compare predictive performance before analysing and investigate whether prediction behaviour changes across demographic groups

over time as well as how the models make their predictions using explainability techniques.

6.1 Comparison of the Models

Table 1 summarises the predictive performance of Logistic Regression, Random Forest and XGBoost at both temporal stages.

Table 1: Global Predictive Performance Metrics Across Temporal Stages

Stage	Model	Acc.	B. Acc.	Prec.	Rec.	F1	AUC
Early (25%)	Log. Reg.	0.7865	0.7890	0.8351	0.7408	0.7851	0.8750
	Rand. Forest	0.8111	0.8138	0.8637	0.7613	0.8093	0.8932
	XGBoost	0.8122	0.8143	0.8548	0.7748	0.8129	0.8950
Full (100%)	Log. Reg.	0.9240	0.9248	0.9451	0.9083	0.9264	0.9747
	Rand. Forest	0.9432	0.9449	0.9763	0.9144	0.9443	0.9839
	XGBoost	0.9484	0.9495	0.9726	0.9281	0.9499	0.9863

As shown in Table 1, all three models demonstrate strong predictive capability using only the first 25% of student activity, indicating that meaningful early identification of at-risk students is feasible. Among the evaluated classifiers, XGBoost achieves the highest performance during the early stage, obtaining a balanced accuracy of 0.8143, an F1-score of 0.8129 and a ROC-AUC of 0.8950, closely followed by Random Forest. Logistic Regression consistently records lower performance across all evaluation metrics, reflecting the limitations of a linear decision boundary when modelling complex student learning behaviour.

Using complete course information substantially improves prediction performance for every classifier. XGBoost remains the best-performing model, achieving an accuracy of 0.9484, balanced accuracy of 0.9495, precision of 0.9726, recall of 0.9281, F1-score of 0.9499 and a ROC-AUC of 0.9863. Random Forest also demonstrates excellent predictive performance, whereas Logistic Regression remains consistently below the ensemble-based approaches.

Overall, these results demonstrate that accurate predictions are already possible after observing only the first quarter of student interactions, but that incorporating additional behavioural and assessment information throughout the course substantially improves the identification of at-risk students. The consistently superior performance of XGBoost across the evaluation metrics motivates its use as the main model for interpreting the fairness and explainability results.

6.2 Fairness Results

To evaluate whether predictive performance is distributed equitably across student populations, fairness was assessed using two complementary metrics: the Selection Rate Gap and the False Negative Rate (FNR) Gap. These metrics were computed across three sensitive attributes: gender, age_band, and the Index of Multiple Deprivation (imd_band).

Within an educational Early Warning System, false negatives represent the most critical prediction error because students incorrectly classified as not at risk receive no academic intervention. Consequently, the FNR gap serves as an important indicator of

²Feature engineering and merging logic: https://github.com/Vlad-Andres/HCMML-project/blob/main/src/feature_engineering.py

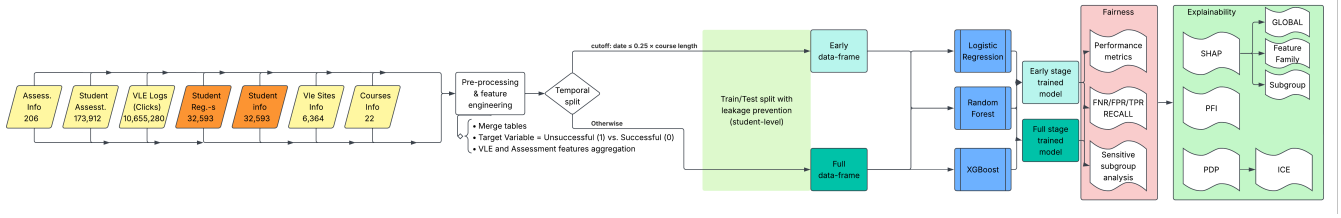


Figure 1: Representation of the full project pipeline described by abstracted processes and states.

whether particular demographic groups are systematically overlooked by the prediction models.

Figure 2 compares the selection-rate gaps for all three classifiers across both temporal stages. The results consistently show that socioeconomic status, represented by the IMD (deprivation) band, exhibits the largest disparity regardless of the underlying model. During the early-course stage, Logistic Regression records the highest IMD selection-rate gap (0.3835), while Random Forest and XGBoost reduce this disparity to approximately 0.31–0.33. Age-related disparities are substantially smaller, whereas gender exhibits the lowest selection-rate differences throughout the study.

Although complete course information reduces disparities across all sensitive attributes, the selection-rate gap associated with IMD remains noticeably larger than those observed for age and gender. This suggests that underlying socioeconomic differences continue to influence which students are classified as at risk even after predictive performance has substantially improved.

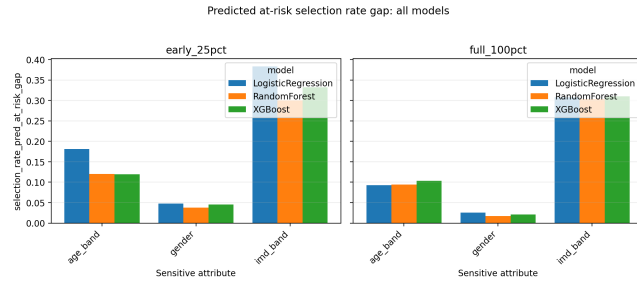


Figure 2: Selection-rate gaps across sensitive demographic attributes for Logistic Regression, Random Forest and XGBoost at the early-course (25%) and full-course (100%) prediction stages.

Figure 3 presents the corresponding False Negative Rate gaps. Compared with the selection-rate analysis, a clear temporal convergence is observed. Across every classifier, FNR disparities decrease substantially as additional behavioural and assessment information becomes available.

The largest improvements occur for age and IMD. For example, the age-based FNR gap of Logistic Regression decreases from approximately 0.19 at the early stage to approximately 0.06 at course completion, while XGBoost reduces the same disparity from approximately 0.06 to 0.03. Similarly, the IMD FNR gap decreases considerably across all three models, with XGBoost achieving the smallest final disparity (approximately 0.056). Gender exhibits the

smallest FNR differences throughout both temporal stages, indicating relatively consistent recall across male and female students.

These observations highlight an important distinction between predictive performance and predictive fairness. While overall accuracy and balanced accuracy improve substantially as more student information becomes available, fairness improvements are not uniform across demographic groups. Error disparities converge considerably over time, particularly for False Negative Rates, yet persistent socioeconomic selection-rate differences remain. Consequently, evaluating fairness only after the completion of a course may conceal important disparities that exist when early-warning interventions are actually deployed.

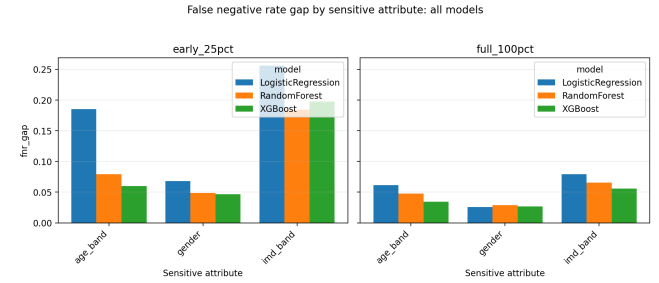


Figure 3: False Negative Rate (FNR) gaps across sensitive demographic attributes for Logistic Regression, Random Forest and XGBoost at the early-course (25%) and full-course (100%) prediction stages.

Overall, the combined performance, fairness, and explainability analyses provide evidence for *partial temporal fairness* convergence. As additional behavioural and assessment information becomes available, predictive performance improves substantially, feature importance shifts from behavioural engagement towards cumulative academic achievement, and demographic error disparities become smaller.

Nevertheless, the persistent socioeconomic disparities observed in both selection-rate and FNR analyses indicate that structural inequalities remain embedded within the learning process, reinforcing the importance of evaluating fairness throughout the prediction timeline rather than solely at the end of a course.

6.3 Explainability Analysis

To understand how prediction behaviour evolves throughout the course, we analysed the trained models using SHAP, Permutation Feature Importance (PFI), and Individual Conditional Expectation

(ICE) plots. Together, these methods provide complementary perspectives on global feature importance, model sensitivity, and individual prediction behaviour.

Figure 4 compares the mean absolute SHAP values across all three models at the early and full-course stages. During the early stage, the models rely primarily on a combination of assessment performance and behavioural engagement features. In particular, cumulative assessment scores, assessment completion, maximum assessment performance, and student engagement within the Virtual Learning Environment (VLE) contribute substantially to model predictions. Since only limited assessment information is available at this stage, behavioural engagement plays an important role in distinguishing between at-risk and successful students.

As additional course information becomes available, the importance of cumulative assessment variables increases considerably. Figure 4(b) shows that cumulative assessment performance and assessment completion become the dominant predictors across all three models, while behavioural engagement variables contribute relatively less. This transition indicates that early predictions depend primarily on student engagement, whereas later predictions are driven largely by accumulated academic performance.

The SHAP observations are further supported by model-agnostic Permutation Feature Importance (PFI). Across both temporal stages, cumulative assessment variables caused the largest decrease in balanced accuracy when permuted, especially features related to weighted assessment scores and assessment completion. We therefore use PFI as a robustness check for the SHAP-based interpretation, but omit the full PFI figure for space.

As a supplementary diagnostic, we also inspected PDP/ICE plots for key engagement features. These plots broadly supported the SHAP and PFI findings: engagement features have stronger effects in the early-course model, while accumulated assessment features dominate in the full-course setting. Since these plots were used only as secondary validation, we omit them from the main report for space; their details and output can be found on GitHub.

7 DISCUSSION

Our results show that machine learning-based Early-Warning Systems (EWS) can identify at-risk students using behavioural and assessment data from OULAD, but that both predictive performance and fairness depend strongly on when the model is evaluated. Even when restricted to the first quarter (25%) of the course timeline, all three models achieve useful predictive performance. XGBoost obtains a balanced accuracy of 0.8143 and an F1-score of 0.8129, suggesting that early intervention is technically feasible before the end of the course. However, the full-course setting produces substantially stronger results, with XGBoost reaching a balanced accuracy of 0.9495 and an at-risk recall of 0.9281. This confirms that later assessment and engagement information provides much stronger predictive signals than the sparse evidence available early in the course.

The fairness results show that improved predictive performance does not automatically imply equal reliability across subgroups. In the early-course setting, false negative rate gaps are larger, especially across age band and `imd_band`. This is important because

false negatives correspond to students who eventually fail or withdraw but are not flagged as at risk, meaning they may miss timely support. For example, the XGBoost `imd_band` FNR gap decreases from 0.1974 in the early-course setting to 0.0556 in the full-course setting, as shown in Figure 3. Similarly, selection-rate disparities remain visible across socioeconomic groups, with an early-stage XGBoost `imd_band` selection-rate gap of 0.3323 in Figure 2. These results support the idea of *partial temporal fairness convergence*: subgroup error disparities shrink as more course information becomes available, but fairness gaps are most pronounced at the stage when intervention would be most useful.

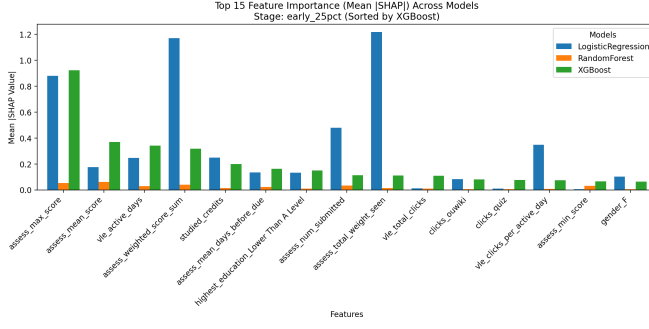
The explainability analysis helps clarify this temporal pattern. In the early-course setting, the available evidence is sparse, so models rely more on a mixture of early assessment signals, engagement behaviour, and static background features. As students progress through the course, cumulative assessment information becomes more complete and increasingly dominates the prediction. This is reflected in the SHAP results: by the full-course setting, assessment-related features become the strongest contributors, while the relative importance of demographic features decreases. Rather than showing that the model actively “removes” bias, this indicates that the model’s decision process shifts from sparse early signals toward more direct evidence of academic progress. Figure 4 illustrates this shift toward accumulated assessment and engagement information.

At the same time, selection-rate gaps remain more persistent than FNR gaps. This suggests that the models continue to reflect underlying differences in observed risk distributions across groups, especially socioeconomic groups represented by `imd_band`. These differences should not be interpreted simply as model bias. Lower IMD (deprivation) bands may show higher observed at-risk rates because of broader structural factors, such as unequal access to learning resources, time constraints, or prior educational opportunities. Therefore, our results distinguish between two related but different phenomena: error-rate disparities, which indicate unequal model reliability, and selection-rate disparities, which may partly reflect real group-level differences in observed educational risk.

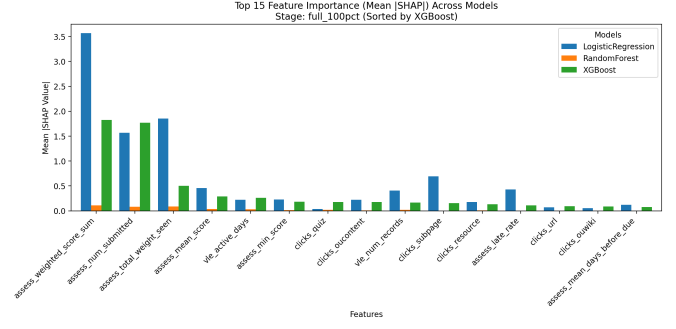
This creates an operational trade-off for universities. A full-course model is more accurate and shows smaller FNR gaps, but it is too late to support students effectively. Conversely, an early-course model is more useful for intervention, but it exhibits larger subgroup reliability gaps. Institutions should therefore avoid treating early-warning systems as fully automated decision-makers. Instead, these systems should be used as decision-support tools, combined with human academic advising and regular subgroup-level audits at the time points when interventions are actually deployed.

7.1 Limitations and Future Improvements

Several limitations remain. First, our analysis uses only two temporal checkpoints: the first 25% of the course and the full-course setting. This captures the contrast between early and complete information, but it does not show how fairness evolves week by week. Future work should evaluate additional checkpoints to identify when subgroup gaps emerge or begin to close. Second, OULAD represents distance-learning courses from a single open-enrolment institution, so the results may not generalize directly to traditional universities or courses with different attendance and assessment



(a) Early-course (25%)



(b) Full-course (100%)

Figure 4: Comparison of mean absolute SHAP values across all three classifiers between the early and full-course prediction stages.

structures. Third, our group fairness analysis relies on categorical subgroup partitioning, which can mask intersectional effects across age, gender, and socioeconomic background. Finally, SHAP, PFI, PDP, and ICE explain model behaviour, but they do not establish causal mechanisms behind student outcomes.

Future work could extend this study by evaluating more temporal checkpoints, testing robustness across different course modules, and exploring intervention policies that combine model alerts with human-led support. Methodologically, future work could also examine explicit fairness constraints or group-specific thresholding strategies, but these should be evaluated carefully to avoid reducing students to demographic categories. For the explainability analysis, a further improvement would be to normalize SHAP magnitudes across models or feature families, since large differences in averaged SHAP values can make direct cross-model comparisons difficult, as suggested by Figure 4.

8 ETHICAL CONSIDERATIONS

Educational early-warning systems can affect how academic support is allocated, so their errors have practical consequences. Historical educational records are not neutral: assessment performance, VLE engagement, and socioeconomic indicators may reflect unequal access to learning resources, time, digital infrastructure, and prior preparation. Models trained on such data may therefore reproduce existing structural disparities even when sensitive attributes are not the main drivers of prediction.

Our results show that fairness should be evaluated at the point of deployment, not only after the course is complete. A model that appears more accurate and fairer in the full-course setting may still show larger subgroup error gaps early in the course, when intervention is most useful. False negatives are especially important because they represent at-risk students who are not flagged for support.

For this reason, early-warning systems should be used as decision-support tools rather than fully automated decision-makers. Institutions should audit subgroup-level error rates, provide explanations to advisors and students where appropriate, and combine model alerts with human-centered academic support.

9 CONTRIBUTIONS

Table 2: Project member contributions

Member	Contribution
Yuxuan Wang	Dataset background research; data pre-processing; implementation of the three models; generation of corresponding plots and fairness analysis code; poster appearance and proofreading; report proofreading and pruning.
Vlad Mutruc	Partial literature review, Project refactoring through architecture modularization, repository setup and maintenance, multi-model combined plots, Poster (structure, methodology diagram and plots), Report (Methods, Model Architecture report sections).
Harshavardhani Suresh Kumar	Literature review support, writing the Abstract and Introduction sections, experiment checking, poster preparation, and presentation preparation.
Kumkum Singh Kumari	Discussions and planning, debugging, testing, poster preparation and presentation, and writing the Results section.

10 AI STATEMENT

Table 3: Use of AI tools

Member	AI tool usage
Yuxuan Wang	Used ChatGPT for brainstorming, code debugging, result interpretation, grammar checking, and LaTeX formatting.
Vlad Mutruc	Grammar checking, code debugging, LaTeX formatting of papers citations.
Harshavardhani Suresh Kumar	Used AI tools for concept clarification, grammar checking, and report wording support.
Kumkum Singh Kumari	Explanation of concepts, grammar checking.

11 CODE REPOSITORY

The code for this project is available at: <https://github.com/Vlad-Andres/HCML-project>

12 CONCLUSION

This project examined temporal fairness in educational early-warning systems using OULAD. We compared an early-course setting using the first 25% of course activity with a full-course setting using all available information. Across three models (Logistic Regression, Random Forest, and XGBoost), predictive performance improved substantially over time, with XGBoost increasing from 0.895 to 0.986 ROC-AUC. This shows that early prediction is feasible, but that later assessment and engagement information provides much stronger predictive signals.

Our fairness analysis shows that accuracy alone is insufficient for evaluating early-warning models. False negative rate gaps were larger in the early-course setting, especially across age band and IMD (deprivation) band, meaning that some groups of at-risk students were more likely to be missed when intervention would be most useful. These gaps decreased with more course information, but selection-rate gaps remained more persistent, suggesting that models still reflect underlying group-level differences in observed risk. SHAP and PFI analyses suggest that predictions are driven primarily by assessment and VLE engagement features, with smaller direct contributions from demographic features. Overall, educational early-warning systems should be audited not only for how accurate they are, but also for when and for whom their predictions become reliable.

REFERENCES

- [1] André Altmann, Laura Tološi, Oliver Sander, and Thomas Lengauer. 2010. Permutation importance: a corrected feature importance measure. *Bioinformatics* 26, 10 (2010), 1340–1347. doi:10.1093/bioinformatics/btq134
- [2] Solon Barocas, Moritz Hardt, and Arvind Narayanan. 2019. *Fairness and Machine Learning: Limitations and Opportunities*. fairmlbook.org. <https://fairmlbook.org/>
- [3] Leo Breiman. 2001. Random forests. *Machine Learning* 45, 1 (2001), 5–32. doi:10.1023/A:1010933404324
- [4] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 785–794. doi:10.1145/2939672.2939785
- [5] Jerome H. Friedman. 2001. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics* 29, 5 (2001), 1189–1232. doi:10.1214/aos/1013203451
- [6] Josh Gardner, Christopher Brooks, and Ryan Baker. 2019. Evaluating the fairness of predictive student models through slicing analysis. In *Proceedings of the 9th International Conference on Learning Analytics & Knowledge*. 225–234. doi:10.1145/3303772.3303791
- [7] Dragan Gašević, Shane Dawson, and George Siemens. 2015. Let’s not forget: Learning analytics are about learning. *TechTrends* 59, 1 (2015), 64–71. doi:10.1007/s11528-014-0822-x
- [8] Alex Goldstein, Adam Kapelner, Justin Bleich, and Emil Pitkin. 2015. Peeking inside the black box: Visualizing statistical learning with plots of individual conditional expectation. *Journal of Computational and Graphical Statistics* 24, 1 (2015), 44–65. doi:10.1080/10618600.2014.907095
- [9] Moritz Hardt, Eric Price, and Nathan Srebro. 2016. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems*, Vol. 29.
- [10] René F. Kizilcec and Hansol Lee. 2020. Algorithmic fairness in education. *arXiv preprint arXiv:2007.05443* (2020).
- [11] Jakub Kuzilek, Martin Hlosta, and Zdenek Zdrahal. 2017. Open University Learning Analytics Dataset. *Scientific Data* 4, 1 (2017), 170171. doi:10.1038/sdata.2017.171
- [12] Scott M. Lundberg and Su-In Lee. 2017. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, Vol. 30.
- [13] Cristóbal Romero and Sebastián Ventura. 2010. Educational data mining: A review of the state of the art. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)* 40, 6 (2010), 601–618. doi:10.1109/TSMCC.2010.2053532
- [14] Renzhe Yu, Hansol Lee, and René F. Kizilcec. 2021. Should college dropout prediction models include protected attributes?. In *Proceedings of the Eighth ACM Conference on Learning @ Scale*. 91–100. doi:10.1145/3430895.3460139